Advanced Machine Learning Framework for Precision Rainfall Prediction for Jharkhand, India

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***Abstract* - Jharkhand, characterized by a substantial agricultural population predominantly reliant on rain-fed agriculture, faces significant challenges due to the erratic nature of precipitation. The study uses meteorological variables and historical rainfall data from the Jharkhand Space Agency Centre (JSAC) to predict rainfall with precision and resilience. Three supervised machine learning algorithms, Random Forest, K-Nearest Neighbour (KNN), and Ridge Regression, are employed to evaluate their performance across monsoon and non-monsoon periods. A novel algorithm is proposed for Jharkhand, offering better modularity and accuracy to predict the Rain Index. The results show the efficacy of these algorithms in capturing the temporal variability of rainfall in Jharkhand. The ensemble modeling model obtained an MSE score of 0.457, providing valuable insights into the viability and competence of machine learning algorithms for rainfall estimation. This research offers a valuable scope for policymakers, researchers, and stakeholders to formulate sustainable strategies to address climate variability and its impact on rain-fed agriculture. The study contributes significantly to meteorological research and offers valuable insights for policymakers, researchers, and stakeholders.**

Keywords – Phase Expansion, Crop Yield, Precipitation Index Prediction, Ensembled

I. INTRODUCTION

Jharkhand, located in eastern India, is known for its diverse topographical landscape and agriculture, which is a central pillar of the state's socioeconomic structure. The state experiences a subtropical climate with seasonal transitions like summer, monsoon, and winter, influenced by topographical gradients, latitudinal influences, and interactions with the Bay of Bengal. The scorching summer months spanning April to June manifest ambient temperatures in the range of 35°C-40°C, characterized by arid air currents and scorching zephyrs. With the advent of the monsoon, spanning the months of June through September demonstrated in Figure 1, the region experiences a deluge of 1,000-1,200 mm of precipitation [8].

This seasonal inundation bears profound significance as a critical resource for sustaining water reservoirs, nurturing soil substrates, and nourishing crop biomass [9]. However, an excess of monsoonal precipitation may precipitate deleterious consequences for agricultural pursuits [10]. The winter season, spanning from November to February, brings milder temperatures within the range of 10°C - 20°C, fostering an environment conducive to crop cultivation by ameliorating moisture stress factors [11]. The geographical heterogeneity ingrained within Jharkhand engenders spatially diverse thermal and precipitation regimes. However, the agrarian sector in Jharkhand is intrinsically susceptible to meteorological variations, notably the dynamics of precipitation patterns, which wield substantial influence over crop yields [12]. Recent years have witnessed heightened rainfall variability due to climate change, underscoring the urgent need for precise and timely precipitation prediction models[13]. Conventional meteorological models, based on physical principles and historical weather data, have historically been the prevailing approach to precipitation prediction. However, these models often fail to capture the complex relationships between meteorological variables and precipitation patterns. Machine learning techniques have emerged as a complementary approach to traditional models, enhancing the accuracy of precipitation forecasts [7]. This research aims to use machine learning techniques to forecast precipitation in Jharkhand using real-time data from Ranchi, focusing on Ranchi as a key consideration.

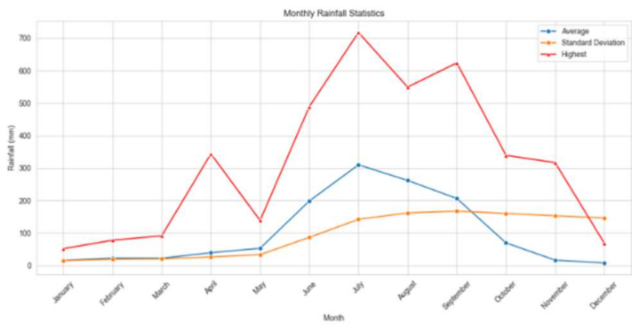


Figure 1 . Visuals of the Statistics of Rain Distribution in Jharkhand Month-Wise for the time span of (1975 – 2020)

The study used machine learning algorithms like Random Forest, Ridge Regression, K-Nearest Neighbours (KNN), and XGBoost to predict rainfall [5]. Random Forest showed the most accurate prediction at 95% accuracy, outperforming other models. The model was trained on a dataset of 22 lakh data points and tested for 15 days in July.

The predictions closely matched actual rainfall events, proving the model's accuracy and practical utility. The goal is to create a predictive framework that explains the relationship between precipitation amounts, soil moisture content, and agricultural production outcomes. This research aims to understand the connections between meteorological forces and crop yields, providing insights to improve agricultural resilience in the face of changing climatic conditions.

This relationship and results from this model were used to “Expanding Crop Yield Data Phases & Optimizing Hyperparameters for Predicting Crop Yield in Jharkhand using Weather Parameters”. Although here in this work only Precipitation Forecasting is focused, whereas the relationship introduced between Rain Index and Crop Yields is shown in another Research Work by the Authors.

II. LITERATURE REVIEW

J. L. Hatfield et al. (2011) created an extensive review of agricultural studies detailing climate change effects alongside adaptive approaches and crop sensitivity. Food protection and security demands immediate mitigation actions according to their study. [1]

U. Verma et al. (2016) established that the best mustard yield prediction accuracy for Haryana India occurs through analysis that combines maximum and minimum temperatures alongside rainfall and relative humidity data and sunshine hours and crop condition information. By reviewing mustard yield factors this study offers critical information [2]. The results from Christopher A. Davis et al.'s 2003 study demonstrated Numerical Weather Prediction models show better accuracy in predicting latitude although propagation accuracy remains challenging thus rainfall statistics may assist NWP predictions for improved forecasting. [3]

Megha Goyal and Urmil Verma (2018) demonstrate that the zonal yield model offers precise and accurate predictions, emphasizing the importance of zonal weather models [6]. An ensemble rainfall forecasting model for Singapore was built by Shan He et al. through their integration of numerical weather prediction with radar-based now-casting models in 2013. By integrating WRF and TM systems this model demonstrates superior performance while significantly advancing meteorology and hydrology practices as well as expanding research related to ensemble rainfall forecasting. [4]

III. METHODOLOGY

The workflow structure and implementations of the study have been mentioned below in the sections discussed. The experimentation conducted for this research study aimed at finding a relationship between the meteorological parameters with the historical crop yields of the state. For this analysis to work the study was distributed through three phases, at first precipitation indexing analysis and prediction for a specific day with respective daily feeds was conducted, then a forward time series forecasting was implemented to predict the weather parameters of the entire year, after that acquiring those data these has to piped lined to the model for crop yield prediction, shown in Figure 2.

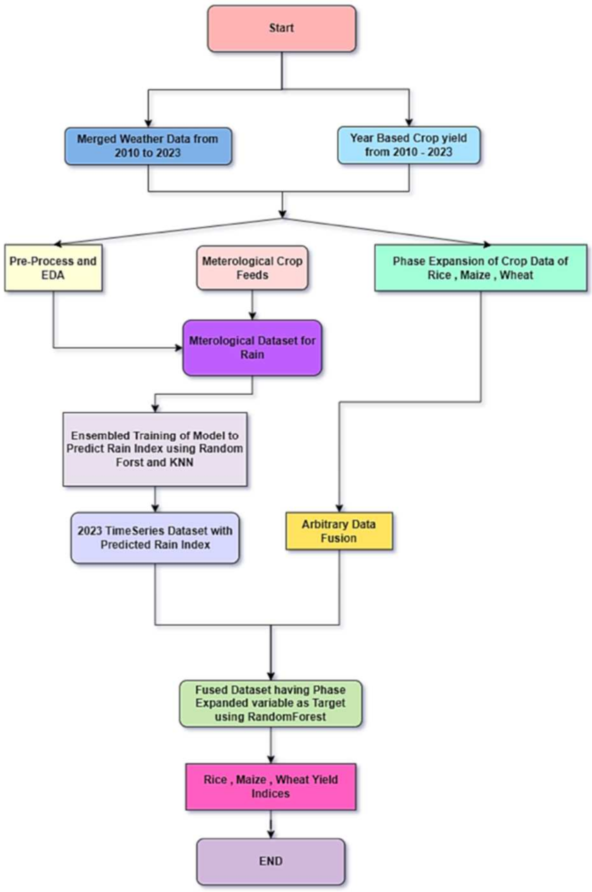


Figure 2 . Experimental Setup

a) Data Acquisition

The experiment used data from 2010 to 2023 from the Jharkhand Space Application Centre (JSAC) for metrological factors. The data, gathered from various observation stations across Jharkhand, included information at specific points of day and intervals. The total count of entries was 22,000, which was modified in three variations for the study phases. The data was private, bringing novelty to the modeling work exclusive to the Jharkhand region.

b) Data Analysis

The study analyzed a dataset containing entries like observation center, latitude, longitude, altitude, high and low temperatures. Table 1 list important entities, but other

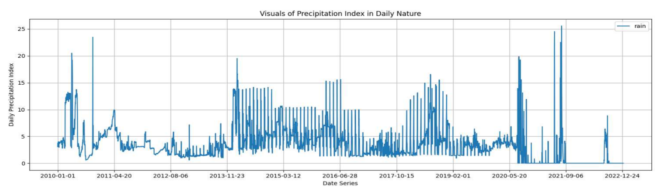


Figure 3. Rainfall Index in mm for Jharkhand over the Years

Table 1 . Description of the Dataset

	count	mean	std	min	max
LATITUDE	2266662	23.0145	3.889858	0	25.036667
LONGITUDE	2266662	82.818518	13.900556	0	87.83667
ALTITUDE(m)	2266662	629	0	629	629
TIME(GMT)	2266662	57.034185	264.076213	0	2345
AIR_TEMPERATURE(deg C)	2266662	24.173096	148.589538	-40	9999.9
RELATIVE_HUMIDITY (%)	2266662	89.033624	532.32464	0	9999.9
WIND_SPEED(m/s)	2266662	0.851926	52.945936	0	9999.9
ATMOSPHERIC_PRESSURE(hPa)	2266662	944.810795	108.254353	0	9999.9
RAIN_DAILY (mm)	2266662	305.572001	327.388429	0	18287
WIND_DIRECTION_AVG(deg)	2266662	348.90431	998.49568	0	9999.9
SUNSHINE_HOUR(minutes)	2266662	429.362161	7998.393496	0	597840

insignificant ones like min and max air temperature and wind direction were not considered. Principal Component Analysis was used to reduce feature dimensionality, as the trend of the data was too valuable to be lost during the reduction process for machine learning model training. Individual features were also a point of interest during analysis.

Table 1. Description of Meteorological Entities

Categories of Features	Description
LATITUDE	The latitude point of the location of observation
LONGITUDE	Longitude point of the location of observation
ALTITUDE(m)	Altitude point of the location of observation
TIME(GMT)	Point of time the observation was recorded in GMT format
AIR_TEMPERATURE(deg C)	Temperature of the atmosphere of the location in Celsius
RELATIVE_HUMIDITY (%)	Relative humidity recorded for the point of observation in percentage format
WIND_SPEED(m/s)	Intensity of Wind recorded for the point of observation in m/s
ATMOSPHERIC_PRESSURE(hPa)	Recorded atmospheric pressure in hPa
RAIN_DAILY(mm)	Precipitation Index (Rainfall Index) of the recorded location

Weather observations are a crucial part of our daily lives, encompassing various factors such as latitude, longitude, time, air temperature, relative humidity, wind speed, atmospheric pressure, daily rainfall, average wind direction, and sunlight hour. These observations are gathered from various locations, with latitudes ranging from 23.0145 to 25.0367 degrees north and longitudes from 82.8185 to 87.8367 degrees east. The range of elevations ranges from 629 to 629 meters. Time measurements are expressed in GMT, with a mean of 24.17 degrees Celsius and a range of –40 to 99.9 degrees Celsius. The mean of relative humidity is 89.03%, while wind speed is 0.85 meters per second. Air pressure measurements range from 944.81 hectopascals to 999.9 hectopascals. The mean of rainfall is 305.57 millimeters, with a daily measurement range of 0 to 18287 millimeters. The average wind direction is 348.9 degrees, and the mean of sunlight hour is 429.36 minutes.

The Rainfall trend from the year 2010 to 2022 is shown in Figure 3, which is the visuals from the newly acquired dataset. Where there was a variable trend between 2010 and 2013, but from 2013 to 2019 showcases an almost constant trend of rainfall. The year of 2016 in Jharkhand experienced a drought situation as per the reports which is visible in the trends [14] [15]. Whereas from 2020 to 2021 an increased hike in rainfall giving the fact that of lockdown due to Lockdowns of the entire nation, which impacted the nature and environment which favoured quality of precipitation whereas from 2022 a decrease is viewed where the possible reason could have been the fact that the Lockdowns were lifted [16].

The data testing process involved removing outliers and performing Multivariate Analysis of Variance (MANOVA) to identify dependencies on the factor of RAIN_DAILY (mm). The mean was then analysed using a weighted linear combination or composite variable. Non-dependant factors like altitudes and aws-index number were dropped during the testing phase. The final dataset was used for modelling the Precipitation Prediction model, an ensemble model of RandomForest Regressor and KNN Regressor. Time-series forecasting was performed for the parameters of the Precipitation Index, but the Precipitation Index did not participate in the ARIMA model's training process. The trained model was deployed for the task. The data testing was performed for the first phase, and non-dependant factors were dropped for the other phases.

c) System and Memory Considerations

This section deals with the system and memory impacts of the described system. Due to the fact that the processes of Rain Index prediction, as well as crop yield estimation, were computationally demanding, the study employed high-performance computing facilities. Memory management was crucial because simple mistakes such as "Memory Overflow" were experienced during early runs. To counter these constraints, optimized hardware configurations were leveraged.

IV. IMPLEMENTATION & EXPERIMENTATION

a) Data Preparation and Preprocessing

The first set of data was obtained from different meteorological observation stations in the state of Jharkhand and included latitude, longitude, height, minimum and maximum air temperature, daily precipitation amounts, relative humidity, air pressure, wind velocity and hours of sunshine. There was temporal variation in the data collected and accumulated data analysis was able to capture trends in the weather impact on crop yield. In the data preparation stage, they were cleaned to reduce inconsistency and redundant records as a way of improving their quality and reliability. For example, those data points that were judged to be biologically or meteorologically impossible or extremely unlikely were excluded from the analysis. Feature selection was then conducted to know which of the parameters was very important in crop yield forecast. To select the best features, Principal Component Analysis (PCA) was used to mitigate the dimensionality of the data while retaining the most fringe complexity. It meant that the proper trends of the data were preserved during model training while reducing the fit of the model to a degree where overfitting occurred [17].

The final dataset was split into training and testing sets in a 50:50 ratio for the initial experimentation phase. Once a suitable model was selected, the entire dataset was used to train the final model, which was then evaluated on real-time data to ensure its robustness.

b) Selection of Machine Learning Models

For the first phase of experimentation, four machine learning models were considered: **Ridge Regression**, **Random Forest Regressor**, **K-Nearest Neighbors (KNN)**, and **XGBoost**. The purpose of the second phase was to evaluate their performance and determine which one of them could be further fine-tuned and used in production.

Ridge Regression: Ridge Regression is a linear regression model whereby the coefficients are chosen to minimize mean squared error and at the same time, the coefficients cannot be too large since the model imposes an L2 penalty on them. Nevertheless, the first tests indicated only moderate performance, although stronger after the hyperparameter setting (e.g., alpha values). As a result, any further testing using this model was stopped contingently of course more testing with this model was stopped.

$$||y - Xw||^{2}{}_{2} + \alpha * ||w||^{2}{}_{2} \quad (1)$$

K-Nearest Neighbours (KNN): KNN works by associating a data point with the class of its closest neighbours mainly contributing to the majority class. In the present study, the Euclidean distance measure was utilized in estimating the similarity. The number of neighbours to consider, n_neighbours, was limited to 15, consistent with precipitation indices information. KNN had moderate accuracy and computational complexity, which placed it as a potential model for ensemble modelling.

$$d(x_{\text{train},i}, x_{\text{test},i}) = \sqrt{\sum (x_{\text{train}}, i - x_{\text{test}}, i)^2} \quad (2)$$

XGBoost: XGBoost is basically a gradient-boosting algorithm and is among the fastest and most accurate ones. However, this study, was not able to generalize well on the test data set, especially when it experienced the problem of overfitting. Therefore, its application was only restricted to the first period of experimentation.

Random Forest: Random Forest proved to be the most potent model when being implemented individually. By averaging decision trees it addressed non-linearity in the data and eliminated overfitting by taking an average of many trees. Hyperparameters including n_estimators which denotes the number of trees, and the random_state which is a random seed were tuned. For randomness within the tree growing process, the random_state was fixed at 42; this allowed for the selection of the best number of trees, n_estimators=90, which improved accuracy and reduced computation time.

c) Ensemble Modeling

As Random Forest and KNN are structurally very different but equally powerful, their combined usage to form an ensemble system opted to improve the predictive performance. A weighted averaging method was used for combining predictions from the two models, as suggested by [8]:

$$\mu = \frac{\sum_{i=1}^{n} w_i y_i}{\sum_{i=1}^{n} w_i} \quad (3)$$

where w_i represents the weight assigned to the i-th model and y_i denotes the prediction from the i-th model. This ensemble method successfully used the strong point of Random Forest which is its immunity to small variations and the localized sensitivity of KNN to obtain increased accuracy and less error ratios.

To evaluate the model's performance, **Mean Squared Error (MSE)** was employed, calculated as:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{y}_i)^2 \quad (4)$$

where Y_i is the actual crop yield and $\hat{y}_i$ is the predicted yield. The results therefore pointed to the versatility of the ensemble model compared to the standalone models in both accuracy and stability.

After having dealt with the Precipitation Index prediction next up was to forecast the time-series data, for which the ARIMA model was used in Mondal et al. (2014), a more general adjustment of an autoregressive average with motion (ARMA) model yields the (ARIMA) model. The classification of this model type is ARIMA(p,d,q), where p represents the autoregressive portions of the data set, d signifies the integrated portions, q represents the moving average portions, and p,d,q are all non-negative integers.

The trained ARIMA was used in weather prediction for the next seasons before feeding into the ensemble model for cropping season yield estimations. The combination of the two in the form of predictive modelling and forecasting offered an optimal solution in the formulation of agricultural plans and decisions.

V. RESULT ANALYSIS

In this section of the study, the insights and results that were acquired during the experimentation are discussed in a precedential order. Here, the second phase is not discussed it's highlighted for the First and Last phases. The format arrangement this section follows is the distribution of parameters related, to the performance of the tested models along with their scores of performance after the insights of the fully trained model.

The heatmap reveals a significant relationship between various meteorological parameters, such as air temperature, relative humidity, wind direction, atmospheric pressure, daily rainfall, wind speed, and sunlight hour in Figure 4. The heat map shows a negative correlation between air temperature and relative humidity, while a positive correlation exists between wind speed and sunlight hour. The daily amount of rain is positively linked with relative humidity, while wind speed and sunlight hour are adversely correlated. High humidity can cause precipitation and cloud formation, while high humidity can slow downwind speed. The daily amount of rain has a negative correlation with wind speed, while atmospheric pressure and sunlight hour have a positive correlation. Rainfall totals and sunlight hours have a negative correlation with atmospheric pressure, as periods of high pressure are linked to clear skies and little precipitation. The sunshine hour and daily rain are inversely connected, as there is less sunshine when clouds cover the sun.

a) Results for Precipitation Models

The results for the models of RandomForest, KNN , XGBoost, and Ridge Regression on the split dataset for training are described here in Table 3.The dataset's top-performing algorithms include Random Forest, XG-Boost, KNN, and Ridge.

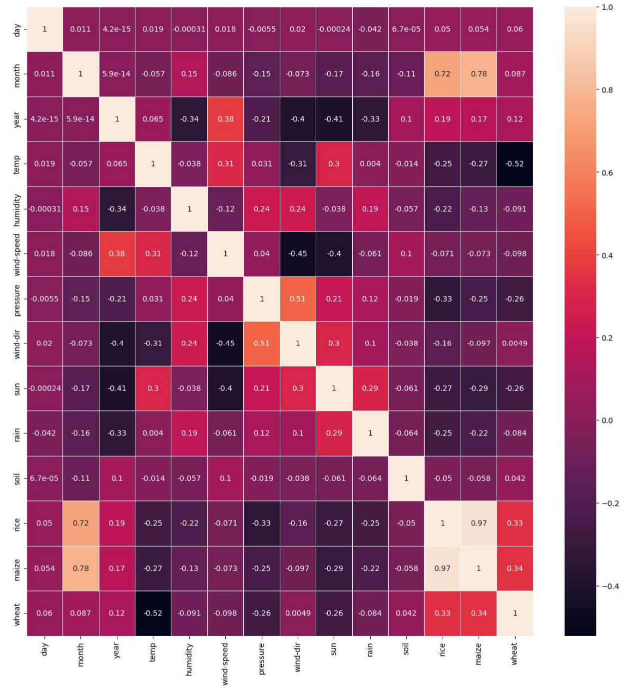


Figure 4. Heatmap visualization of the dataset of First Phase Experimentation

The graph shows a wide variation in the values of these algorithms. RandomForest and XG Boost perform exceptionally well, but XG-Boost is not as effective due to its negative precipitation index predictions. RandomForest and KNN are the most appropriate models for predicting a higher range of values, while KNN's model predicts a lower range of data. An ensemble model of both algorithms is considered wise for optimal performance.

b) Ensemble Model Results for the Entire Dataset (2010–2022)

Having concluded the entirety of the dataset was trained with Random Forest and KNN. It took 46 minutes for Random Forest and 10 minutes for KNN to get trained. Random Forest was acquiring 3gbs worth of space even after using a compression mechanism whereas KNN. So, when the ensembled mechanism was used that is RandomForest with first priority and KNN on Second priority, (1 * 0.41523438549184655 + 2 * 0.03952815582675963) / (1+2) inches 0.15158751377286873 inches This value when checked with the online Realtime feeds service named Ventusky described in Table 9 .

VI. RESULT'S DISCUSSION AND INTERPRETATIONS

Having the Ensembled Precipitation Index prediction model in a stage of procuring and anticipating the unforeseen rain patterns, the prebuilt dataset for 2023 having the predictors of the model similar to this work was the Time Series Forecast with ARIMA model. The series prediction displays the anticipated millimeter-level rain index for the year 2023 in Figure 5. The date is displayed on the x-axis, while the millimeter-based rain index is displayed on the y-axis. The graph indicates that with values over 20 millimeters, August and September are predicted to have the greatest rain index values. February and March should have the lowest rain index readings, with readings fewer than 5 millimeters.

Table 3. Performance of the various Models for the Precipitation Index Prediction

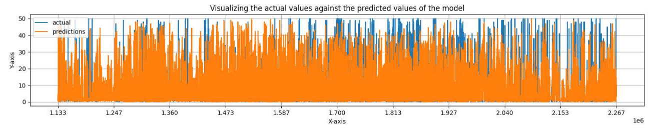
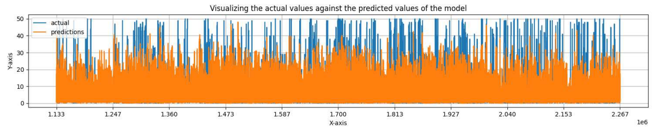
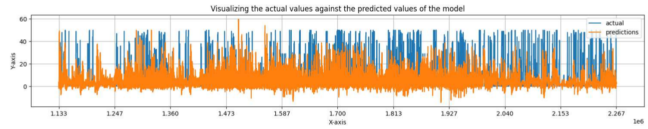
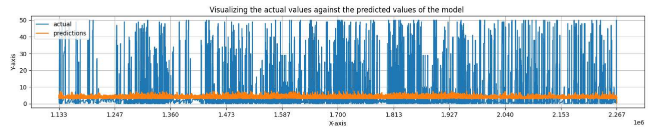
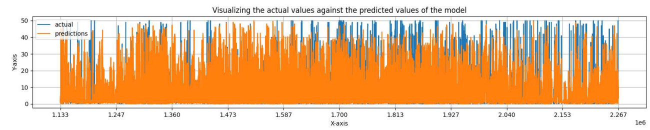
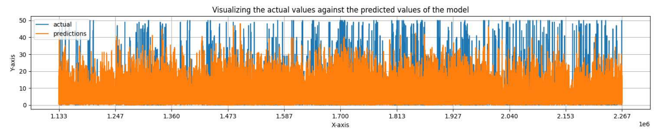
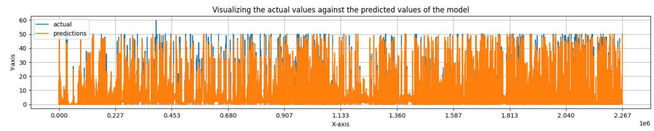
Models	Training Accuracy	MSE Score	Performance Visuals
RandomForest	0.93710217399656	4.992858739581716	
KNN	0.26223673924329	5.2196805425853565	
XGBoost	0.40984956895851	5.2196805425853565	
Ridge Regression	0.01323113188912	5.555334011486265	

Table 4 . Optimal Solution for the Prediction Model for Precipitation

Models	Training Accuracy	MSE Score	Performance Visuals
RandomForest	0.95029451626583	0.7571893938491224	
KNN	0.28093591043475	4.059230504656351	
Ensembled	**0.96119251626482**	**0.45776712598513**	

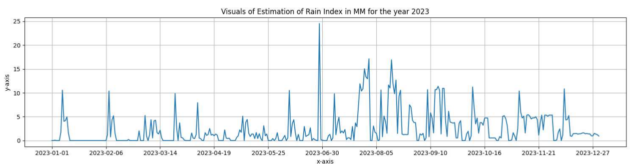


Figure 5 . Visualization of the Estimation of the Precipitation for the Year 2023

Additionally, the graph displays a general tendency for a rising rain index from January to August, and then a falling rain index from September to December. The comparison of the proposed model has been presented in Table 3 where 60–90% training accuracy gap between Random Forest (RF) and the other models arises because RF is a high-capacity ensemble model that can learn complex patterns and often overfits the training data, resulting in very high accuracy. In contrast, models like KNN, XGBoost (likely under-tuned), and Ridge Regression (a simple linear model) lack the same complexity or are poorly optimized, causing them to underfit and yield significantly lower training accuracy on the same dataset.

In table 4, the combination of KNN with RF have smoothed out prediction errors—yielding a similar overall MSE compared to XGBoost—while still suffering from less precise classification boundaries; in contrast, XGBoost's sequential boosting allows it to capture intricate patterns and make more confident class distinctions, thereby achieving a higher accuracy score even when the MSE, which averages error magnitude, remains comparable between the two approaches.

As studied in the work, various machine learning algorithms have been used for a variety of purposes, with varied degrees of accuracy and inaccuracy. In Pakistan, a fusion strategy had an RMSE of ~0.4667, whereas in the US, an

ANN-based hybrid method showed 87% accuracy. In India, feature selection and Bayesian models produced 91% accuracy, while decision-tree-based techniques obtained 72.3%. The accuracy of CART and K-means techniques in the Mahanadi River Basin was 63.7%, while an ANN-based model in Assam was 63%. CART and C4.5 obtained 99.3% accuracy locally, whereas hybrid models such as BLR, BDTR, and others had an RMSE of 0.117 in Malaysia. An SVR-RF model in Andhra Pradesh has RMSE 15.46, but hybrid multi-models in Japan had RMSE 33.42. In Jharkhand, a suggested ensembled Random Forest and KNN model obtained an MSE of 0.45.

CONCLUSION

Concluding with Jharkhand, a region heavily reliant on rain-fed agriculture, faces significant challenges due to erratic precipitation dynamics. Accurate rainfall prediction is crucial for timely planning, risk mitigation, and effective crop-related strategies. Machine learning algorithms, such as Random Forest and K-Nearest Neighbour (KNN), are used to predict rainfall with precision and resilience. The study demonstrates the applicability of machine learning algorithms in capturing the intricate spatiotemporal variability of rainfall in Jharkhand. The ensembled method was used to find and predict the precipitation index, achieving an MSE Score of 0.457. The data was then pipelined to find crop yield data, as proposed in another work. However, the study has some limitations, such as not considering the Pollution Index and recommending proper memory management and optimized systems for the Precipitation Model to work effectively. Future research should explore the use of additional environmental and climatic variables to enhance the precision of the forecasting system.

Declaration Of Availability of Data and Material

The AWS data of metrological factors from 2010 to 2023 (July) were acquired through JSAC (Jharkhand Space Application Center), these data contained information at individual points of day at certain intervals. These data were an amalgamation of different observation stations across Jharkhand.

Acknowledgement

This Project was done in collaboration with AIIT (Amity Institute of Information And Technology) Amity University, Jharkhand and JSAC. The Implementation of the Project Can be found in this GitHub Repo – "*https://github.com/rjzeref/JSAC-Crop-Yield-Prediction*", which will be made public upon publication due to integrity concerns.

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